C

IB MATH: APPLICATIONS & INTERPRETATION • UNIT 7.01

Logistic Regression & Classification

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

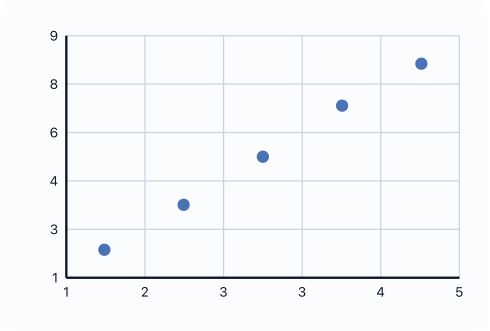
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 A
Q6 A	Q7 A	Q8 A	Q9 C	Q10 C
Q11 D	Q12 B	Q13 B	Q14 A	Q15 D
Q16 B	Q17 C	Q18 C	Q19 C	Q20 B
Q21 A	Q22 D	Q23 C	Q24 A	Q25 C

Logistic Regression & Classification Practice Exercises

1 MCQ

In evaluating a binary classifier, the confusion matrix shows $TP = 45$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.



- **A** 81.0% ✓
- **B** 81.8%
- **C** 52.4%
- **D** 13

WORKED SOLUTION & VERIFICATION

Accuracy is the ratio of correct predictions to total samples: $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{45+40}{105} = \frac{85}{105} = 81.0\%$.

2 MCQ

In evaluating a binary classifier, the confusion matrix shows $TP = 55$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

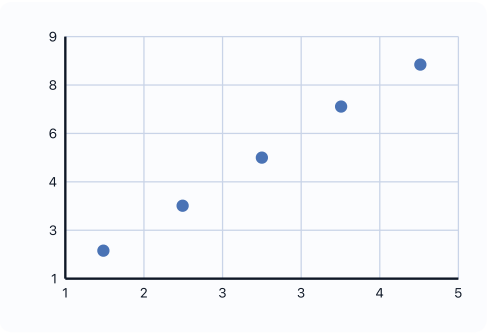
- **A** 82.6% ✓
- **B** 84.6%
- **C** 56.5%
- **D** 13

WORKED SOLUTION & VERIFICATION

Accuracy is the ratio of correct predictions to total samples: $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{55+40}{115} = \frac{95}{115} = 82.6\%$.

3 MCQ

In evaluating a binary classifier, the confusion matrix shows $TP = 60$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.



- **A** 83.3% ✓
- **B** 85.7%
- **C** 58.3%
- **D** 13

WORKED SOLUTION & VERIFICATION

Accuracy is the ratio of correct predictions to total samples: $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{60+40}{120} = \frac{100}{120} = 83.3\%$.

4 MCQ

In evaluating a binary classifier, the confusion matrix shows $TP = 70$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

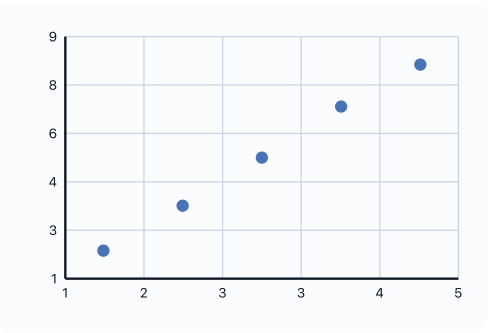
- **A** 84.6% ✓
- **B** 87.5%
- **C** 61.5%
- **D** 13

WORKED SOLUTION & VERIFICATION

Accuracy is the ratio of correct predictions to total samples: $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{70+40}{130} = \frac{110}{130} = 84.6\%$.

5 MCQ

A spam classification model identifies $TP = 30$ genuine spam emails and $FP = 10$ legitimate emails falsely flagged as spam. What is the precision of this model?



- **A** 75.0% ✓
- **B** 60.0%
- **C** 25.0%
- **D** 85.7%

WORKED SOLUTION & VERIFICATION

Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP+FP} = \frac{30}{30+10} = \frac{30}{40} = 75.0\%$.

6 MCQ

A spam classification model identifies $TP = 60$ genuine spam emails and $FP = 20$ legitimate emails falsely flagged as spam. What is the precision of this model?

- **A** 75.0% ✓
- **B** 60.0%
- **C** 25.0%
- **D** 85.7%

WORKED SOLUTION & VERIFICATION

Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP+FP} = \frac{60}{60+20} = \frac{60}{80} = 75.0\%$.

7 MCQ

A spam classification model identifies $TP = 90$ genuine spam emails and $FP = 30$ legitimate emails falsely flagged as spam. What is the precision of this model?

- **A** 75.0% ✓
- **B** 60.0%
- **C** 25.0%
- **D** 85.7%

WORKED SOLUTION & VERIFICATION

Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP+FP} = \frac{90}{90+30} = \frac{90}{120} = 75.0\%$.

8 MCQ

A spam classification model identifies $TP = 120$ genuine spam emails and $FP = 40$ legitimate emails falsely flagged as spam. What is the precision of this model?

- **A** 75.0% ✓
- **B** 60.0%
- **C** 25.0%
- **D** 85.7%

WORKED SOLUTION & VERIFICATION

Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP+FP} = \frac{120}{120+40} = \frac{120}{160} = 75.0\%$.

9 MCQ

Consider a simple logistic regression model where the decision boundary is defined by the linear equation $w_1x_1 + w_2x_2 + b = 0$. If the weights are $w_1 = 2$ and $w_2 = 3$, and the bias is $b = -7$, what is the equation of the decision boundary?

- **A** $3x_1 + 2x_2 + 7 = 0$
- **B** $2x_1 + 3x_2 + 7 = 0$
- **C** $2x_1 + 3x_2 - 7 = 0$ ✓
- **D** $-2x_1 - 3x_2 + 7 = 0$

WORKED SOLUTION & VERIFICATION

The decision boundary in a logistic regression model is defined by the equation where the log-odds are zero. This corresponds to a linear combination of the input features and weights, plus the bias, being equal to zero. Given the weights $w_1 = 2$ and $w_2 = 3$, and the bias $b = -7$, the equation of the decision boundary is $w_1x_1 + w_2x_2 + b = 0$. Substituting the given values, we get $2x_1 + 3x_2 + (-7) = 0$, which simplifies to $2x_1 + 3x_2 - 7 = 0$.

10 MCQ

Consider a simple logistic regression model where the decision boundary is defined by the equation $-1.5x_1 + 2x_2 + 5 = 0$. For a data point (x_1, x_2), if $-1.5x_1 + 2x_2 + 5 < 0$, it is classified into Class 0. Otherwise, it is classified into Class 1. Which of the following points lies on the decision boundary?

- **A** $(-1, 1)$
- **B** $(2, -2)$
- **C** $(0, -2.5)$ ✓
- **D** $(3, 0.5)$

WORKED SOLUTION & VERIFICATION

The decision boundary is defined by the equation $-1.5x_1 + 2x_2 + 5 = 0$. To find which point lies on the boundary, we substitute the coordinates of each point into the equation. For point $(-1, 1)$: $-1.5(-1) + 2(1) + 5 = 1.5 + 2 + 5 = 8.5$, which is not 0. For point $(2, -2)$: $-1.5(2) + 2(-2) + 5 = -3 - 4 + 5 = -2$, which is not 0. For point $(0, -2.5)$: $-1.5(0) + 2(-2.5) + 5 = 0 - 5 + 5 = 0$. This point lies on the boundary. For point $(3, 0.5)$: $-1.5(3) + 2(0.5) + 5 = -4.5 + 1 + 5 = 1.5$, which is not 0. Therefore, the point $(0, -2.5)$ lies on the decision boundary.

11 MCQ

Considering evaluating a binary classifier, the confusion matrix shows $TP = 55$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

Binary Classification Contingency Table

	Pred Positive	Pred Negative
Actual Positive	55	10
Actual Negative	10	40

A 56.5%

B 13

C 84.6%

D 82.6% ✓

WORKED SOLUTION & VERIFICATION

Accuracy is the ratio of correct predictions to total samples: $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{55+40}{115} = \frac{95}{115} = 82.6\%$.

12 MCQ

Examining evaluating a binary classifier, the confusion matrix shows $TP = 60$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

Binary Classification Contingency Table

	Pred Positive	Pred Negative
Actual Positive	60	10
Actual Negative	10	40

A 85.7%

B 83.3% ✓

C 58.3%

D 13

WORKED SOLUTION & VERIFICATION

Accuracy is the ratio of correct predictions to total samples: $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{60+40}{120} = \frac{100}{120} = 83.3\%$.

13 MCQ

For the configuration in evaluating a binary classifier, the confusion matrix shows $TP = 70$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

Binary Classification Contingency Table

	Pred Positive	Pred Negative
Actual Positive	70	10
Actual Negative	10	40

A 87.5%

B 84.6% ✓

C 61.5%

D 13

WORKED SOLUTION & VERIFICATION

Accuracy is the ratio of correct predictions to total samples: $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{70+40}{130} = \frac{110}{130} = 84.6\%$.

14 MCQ

Consider a binary classification problem where the decision boundary is defined by the logistic function $p(x) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2)}}$, which represents the probability of a data point $\mathbf{x} = (x_1, x_2)$ belonging to class 1. The estimated model parameters are $\beta_0 = -3$, $\beta_1 = 1$, and $\beta_2 = -2$. We are interested in the geometric shape of the classification boundary. The classification boundary occurs where $p(x) = 0.5$. Find the equation of this boundary in the x_1x_2-plane.

A $x_2 = \frac{1}{2}x_1 - \frac{3}{2}$ ✓

B $x_2 = 2x_1 - 3$

C $x_2 = 2x_1 + 3$

D $x_2 = \frac{1}{2}x_1 + \frac{3}{2}$

WORKED SOLUTION & VERIFICATION

The logistic regression model gives the probability of belonging to class 1 as $p(x) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2)}}$. The classification boundary is the set of points where $p(x) = 0.5$.

Setting $p(x) = 0.5$, we have:

$0.5 = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2)}}$

This implies:

$1 + e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2)} = 2$

Therefore:

$e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2)} = 1$

Taking the natural logarithm of both sides:

$-(\beta_0 + \beta_1 x_1 + \beta_2 x_2) = \ln(1)$

Since $\ln(1) = 0$, we get:

$\beta_0 + \beta_1 x_1 + \beta_2 x_2 = 0$

This is the equation of the linear decision boundary.

Substitute the given parameter values: $\beta_0 = -3$, $\beta_1 = 1$, and $\beta_2 = -2$.

$-3 + x_1 + (-2) \cdot x_2 = 0$

$-3 + x_1 - 2x_2 = 0$

We need to express this in the form $x_2 = f(x_1)$.

$2x_2 = x_1 - 3$

$x_2 = \frac{1}{2}x_1 - \frac{3}{2}$

This matches choice A.

15 MCQ

Consider a simple logistic regression model where the decision boundary is defined by the equation $3x + 4y - 12 = 0$. This equation represents a linear boundary in a 2D feature space. What is the y-intercept of this decision boundary?

A $y = -4$

B $y = -3$

C $y = 4$

D $y = 3$ ✓

WORKED SOLUTION & VERIFICATION

The equation of the decision boundary is given by $3x + 4y - 12 = 0$. To find the y-intercept, we set $x = 0$ and solve for y. \nSubstituting $x = 0$ into the equation: \n$3(0) + 4y - 12 = 0$

$4y - 12 = 0$

$4y = 12$

$y = \frac{12}{4}$

$y = 3$

Therefore, the y-intercept of the decision boundary is 3.

16 MCQ

A binary classification model uses a logistic sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$. The decision boundary is defined by $\sigma(z) = 0.5$, where $z = w_1x_1 + w_0$. For a dataset with two features x_1 and x_2, the model's linear component is $z = 2x_1 - x_2 + b$. If the classification boundary passes through the point $(x_1, x_2) = (3, 1)$, what is the value of the bias b?

A $b = 7$

B $b = -5$ ✓

C $b = -7$

D $b = 5$

WORKED SOLUTION & VERIFICATION

The logistic sigmoid function is $\sigma(z) = \frac{1}{1+e^{-z}}$. The decision boundary is where $\sigma(z) = 0.5$. Setting $\sigma(z) = 0.5$ gives $\frac{1}{1+e^{-z}} = 0.5$, which implies $1 = 0.5(1 + e^{-z})$, so $2 = 1 + e^{-z}$, and $e^{-z} = 1$. This occurs when $-z = 0$, or $z = 0$.
Therefore, the decision boundary is defined by $z = 0$.
The linear component of the model is given by $z = w_1x_1 + w_2x_2 + b$. In this problem, $z = 2x_1 - x_2 + b$.
Since the decision boundary is $z = 0$, we have $2x_1 - x_2 + b = 0$.
The classification boundary passes through the point $(x_1, x_2) = (3, 1)$. Substituting these values into the equation for the decision boundary:
$2(3) - 1 + b = 0$
$6 - 1 + b = 0$
$5 + b = 0$
$b = -5$

17 MCQ

In an applied problem, a spam classification model identifies $TP = 90$ genuine spam emails and $FP = 30$ legitimate emails falsely flagged as spam. What is the precision of this model?

A 25.0%

B 85.7%

C 75.0% ✓

D 60.0%

WORKED SOLUTION & VERIFICATION

Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP+FP} = \frac{90}{90+30} = \frac{90}{120} = 75.0\%$.

18 MCQ

Suppose a spam classification model identifies $TP = 120$ genuine spam emails and $FP = 40$ legitimate emails falsely flagged as spam. What is the precision of this model?

A 85.7%

B 60.0%

C 75.0% ✓

D 25.0%

WORKED SOLUTION & VERIFICATION

Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP+FP} = \frac{120}{120+40} = \frac{120}{160} = 75.0\%$.

19 MCQ

An observed spam classification model identifies $TP = 150$ genuine spam emails and $FP = 50$ legitimate emails falsely flagged as spam. What is the precision of this model?

- A 60.0%
- B 25.0%
- C 75.0% ✓
- D 85.7%

WORKED SOLUTION & VERIFICATION

Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP+FP} = \frac{150}{150+50} = \frac{150}{200} = 75.0\%$.

20 MCQ

Consider a spam classification model identifies $TP = 180$ genuine spam emails and $FP = 60$ legitimate emails falsely flagged as spam. What is the precision of this model?

- A 25.0%
- B 75.0% ✓
- C 85.7%
- D 60.0%

WORKED SOLUTION & VERIFICATION

Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP+FP} = \frac{180}{180+60} = \frac{180}{240} = 75.0\%$.

21 MCQ

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.30$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

- A -0.847 ✓
- B 0.429
- C 0.353
- D -2.347

WORKED SOLUTION & VERIFICATION

Odds are $\frac{p}{1-p} = \frac{0.30}{0.70} = 0.4286$. Taking natural logarithm: $\ln(0.4286) = -0.847$.

22 MCQ

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.40$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

- A 0.667
- B 0.795
- C -1.905
- D -0.405 ✓

WORKED SOLUTION & VERIFICATION

Odds are $\frac{p}{1-p} = \frac{0.40}{0.60} = 0.6667$. Taking natural logarithm: $\ln(0.6667) = -0.405$.

23 MCQ

In an applied problem, a logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.50$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

A -1.500

B 1.200

C 0.000 ✓

D 1.000

WORKED SOLUTION & VERIFICATION

Odds are $\frac{p}{1-p} = \frac{0.50}{0.50} = 1.0000$. Taking natural logarithm: $\ln(1.0000) = 0.000$.

24 MCQ

Suppose a logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.60$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

A 0.405 ✓

B -1.095

C 1.605

D 1.500

WORKED SOLUTION & VERIFICATION

Odds are $\frac{p}{1-p} = \frac{0.60}{0.40} = 1.5000$. Taking natural logarithm: $\ln(1.5000) = 0.405$.

25 MCQ

An observed logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.70$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

A -0.653

B 2.047

C 0.847 ✓

D 2.333

WORKED SOLUTION & VERIFICATION

Odds are $\frac{p}{1-p} = \frac{0.70}{0.30} = 2.3333$. Taking natural logarithm: $\ln(2.3333) = 0.847$.